

Lakshyaraj Singh Rao

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SUMMARY

I build the **reliability layer that lets AI systems run in production without supervision** — refusal when retrieval fails, abort when output goes off-script, eval-gating before merge, idempotency under webhook re-delivery. Co-Founder and sole engineer on **Homesty.ai LLP** since Nov 2025, serving real traffic on Next.js 15 + pgvector + GPT-4o. **Four npm packages** (one with SLSA provenance) and a **GitHub Marketplace Action** — extracted from what production actually broke.

TECHNICAL SKILLS

LANGUAGES TypeScript · Python · SQL

BACKEND & AI Node.js, Next.js 15, PostgreSQL, pgvector, Prisma 7, Redis, BullMQ, OpenAI, Anthropic, RAG, LLM streaming

INFRASTRUCTURE Docker, Kubernetes, Helm, Terraform, ArgoCD, Argo Rollouts, GitHub Actions, Vercel

OBSERVABILITY Sentry · Prometheus · Grafana

SECURITY HMAC webhooks, prompt-injection defense, idempotency, rate limiting

PRACTICES Idempotent workers, canary rollouts, CI/CD, observability-first design

EXPERIENCE

[Homesty.ai LLP](#) — Co-Founder & Sole Engineer

Nov 2025 — Present

- Sole engineer on a live buyer-side real-estate AI: Next.js 15, Postgres + pgvector, Prisma, GPT-4o and Claude streaming, behind rate limiting, intent routing, and Sentry.
- Built the chat around refusal-first retrieval and a mid-stream guardrail — the AI declines when it has no grounded source rather than improvising, which in real estate is a liability, not a UX choice.
- Chose pgvector over a managed vector DB to keep relational + vector queries in one connection pool — faster joins on entity metadata, simpler ops.
- Hardened those patterns into standalone OSS still running in production: Anchor (provenance RAG) and Tripwire (streaming safety).
- Stack: Next.js 15 · PostgreSQL · pgvector · Prisma 7 · GPT-4o · Claude · Sentry · Upstash Redis · Docker · Vercel.

PROJECTS

Anvil — idempotent webhook → BullMQ pipeline [github](#) · [@ykstormorg/anvil](#)

Stripe sends the same webhook three times when the internet hiccups, and your worker double-charges users. **Anvil is the dedup + retry + dead-letter pipeline that handles every webhook source (Stripe, GitHub, Twilio) so your worker never repeats.** Drop-in library, npm v0.1.0 with build provenance. Ships a Terraform module for production and a Helm chart for Kubernetes.

Anchor — provenance-first RAG [live playground](#) · [github](#)

Every RAG demo breaks when retrieval finds nothing useful — the AI just makes things up. **Anchor is the RAG layer that refuses instead of guessing: above the similarity floor it returns chunks with provenance; below, it returns refused: true.** Built into a real-estate production app where a hallucination is a lawsuit. Live playground, one-command Docker setup.

Tripwire — mid-stream LLM guardrail [github](#) · [@ykstormorg/tripwire](#)

Once an LLM token reaches the user it's already wrong — post-hoc moderation runs after the damage. **Tripwire watches each token mid-stream and kills the response on rule trip: PII leaks, prompt-injection, fabricated facts.** Drop in as an OpenAI-compatible proxy (change one URL) or import as a library. Sub-millisecond per chunk; zero buffering on the happy path.

Goldset — LLM evaluation GitHub Action [github](#) · [marketplace](#) · [@ykstormorg/goldset](#)

"Did my prompt change break anything?" — today you eyeball outputs in a notebook and ship a guess. **Goldset is CI for AI: three eval runners (golden answers, LLM judge, structural assertions) run on every PR, post a delta-vs-base comment, and block merge on regression.** GitHub Marketplace Action + npm library, pluggable OpenAI / Anthropic judge.

Quickdraw — LLM streaming benchmark CLI [github](#) · [@ykstormorg/quickdraw](#)

Every AI provider claims to be "fastest" — but fast on what prompts, what network, what time of day? **Quickdraw measures real LLM streaming on your prompts: time-to-first-token, tokens-per-second, cost-per-1K across OpenAI + Anthropic.** quickdraw diff catches regressions between nightly runs. On npm with SLSA build provenance — verifiable supply chain.

Stackup — Kubernetes GitOps lab [docs](#) · [github](#)

Production Kubernetes patterns cost \$200+/month on cloud, and most "K8s tutorials" stop at kubectl run nginx. **Stackup runs the full production stack on your laptop in ~15 minutes: ArgoCD GitOps app-of-apps, Argo Rollouts canary with real Prometheus success-rate analysis, kube-prometheus-stack monitoring.** Real progressive delivery, free.

Codecraft — in-browser IDE with WebContainers [live demo](#) · [github](#)

"Try-it" links die when users have to install Node, clone the repo, and run npm install. **Codecraft runs a real Vite + React dev server inside the browser tab — edit code in Monaco, type npm install dayjs in the terminal, watch the preview hot-reload in under 2 seconds.** No backend, no install. Cross-origin isolation is the trick; an IndexedDB snapshot cache drops repeat-visit boots under 20s.

EDUCATION

Manipal University Jaipur — B.Tech, Computer Science · Graduating 2026

Coursework: Distributed Systems · Database Internals · Networks · Machine Learning Foundations.

Devagiri CMI Public School, Calicut — CBSE Class XII, 84%

Vidya Kendra, Calicut — CBSE Class X, 94%